

A Process-Based Coherence Field Bridging Classical and Quantum Dynamics (Not-Math Framework)

Joseph Forrester
Independent Researcher
jcfunited@gmail.com

January 2, 2026

Abstract

We present a conservative field framework—the Universal Field of Coherent Processes (UFCP)—in which conventional “numbers” are reinterpreted as processual quantities that carry local tempo, phase, and coherent potential. From a single Hermitian action with symmetric couplings we derive equations of motion, Noether currents, and precise correspondence limits that reproduce (i) the Schrödinger equation and Born’s rule, (ii) Newtonian mechanics via Ehrenfest/collective-coordinate limits, and (iii) relativistic wave dynamics with microcausality in Klein–Gordon/Dirac corners. Measurement probabilities emerge by extremizing captured coherent potential under a conserved norm, uniquely yielding the overlap-squared rule under unitary invariance and orthogonal additivity. The framework makes falsifiable predictions including a coupling-induced tempo dilation of coherently “crowded” oscillators and a coherence-resource trade-off in multi-path/party interferometry. We outline laboratory protocols and list explicit failure conditions (e.g., non-Hermitian kernels, asymmetric interactions) that would falsify the approach. This unifies classical, quantum, and relativistic behaviors as projections of a single process-based dynamics while remaining testable in precision metrology.

1 Introduction

We develop a process-based field description where quantities are not static scalars but *coherent processes* carrying a local tempo ω , phase ϕ , and coherent-potential density ρ . The ambition is pragmatic: recover known laws (classical, quantum, relativistic) from one conservative action while exposing new, falsifiable effects in precision metrology. Our construction is conservative in the Noether sense (global U(1), Poincaré, and gauge symmetries imply conserved currents) and reduces to familiar theories in appropriate limits.

2 Coherence Field & Action

Let $\psi = \sqrt{\rho} e^{i\phi}$ be the coherence field on $(\mathcal{M}, g_{\mu\nu})$ with minimal gauge coupling $D_\mu = \partial_\mu + \frac{iq}{\hbar} A_\mu$. Choose a Hermitian kinetic operator $\hat{\Omega}_g$ (e.g., nonrelativistic $\hat{\Omega}_g = -\frac{\hbar^2}{2m} g^{ij} D_i D_j + V$, or relativistic). With a symmetric interaction kernel W , the action is

$$S = \int d^4x \sqrt{-g} \left[\frac{i\hbar}{2} (\psi^{D_t \psi - \psi D_t \psi}) - \psi^{\hat{\Omega}_g \psi - \mathcal{U}(\rho) - \frac{1}{2} \int \rho(x) W(x-x') \rho(x') \sqrt{-g'} d^4x'} \right].$$

Variation yields the equation of motion

$$i\hbar D_t\psi = \hat{\Omega}_g \psi + \frac{\partial \mathcal{U}}{\partial \rho} \psi + (W * \rho) \psi. \quad (1)$$

Global U(1) symmetry gives $\nabla_\mu J^\mu = 0$ with $J^0 = \rho$ and spatial current $\mathbf{J} = \rho \mathbf{v}$ (for quadratic $\hat{\Omega}_g$: $\mathbf{v} = \frac{\hbar}{m}(\nabla\phi - q\mathbf{A}/\hbar)$). Variation w.r.t. $g^{\mu\nu}$ yields the stress-energy tensor $T_{\mu\nu}$; with an Einstein-Hilbert term this couples to gravity in the standard way.

3 Correspondence Proofs

Schrödinger. For $\hat{\Omega} = -\frac{\hbar^2}{2m}\nabla^2 + V$, Eq. (1) is $i\hbar\partial_t\psi = (-\frac{\hbar^2}{2m}\nabla^2 + V)\psi$. Writing $\psi = \sqrt{\rho}e^{i\phi}$ gives the Madelung form with the quantum potential $Q = -\frac{\hbar^2}{2m}\frac{\nabla^2\sqrt{\rho}}{\sqrt{\rho}}$.

Classical (Newton/Ehrenfest). When the smoothness scale L of ρ satisfies $L \gg \lambda_{dB} = h/(mv)$ and $|\nabla Q| \ll |\nabla V|$, the packet's center obeys $m\ddot{\mathbf{X}} = -\nabla V(\mathbf{X})$.

Relativistic. Choosing a relativistic kinetic operator (Klein-Gordon/Dirac) yields microcausal dynamics and bounded group velocity; minimal coupling preserves gauge currents.

4 Measurement as Extremum (Born Rule)

Let $\{\psi_k\}$ be orthonormal detector modes and $\alpha_k = \langle\psi_k|\psi_S\rangle$. We posit: (i) norm conservation $\sum_k p_k = 1$, (ii) unitary invariance, (iii) orthogonal additivity, (iv) reciprocity (bilinear capture). Maximizing $\sum_k p_k |\alpha_k|^2$ under (i) gives $p_k = |\alpha_k|^2 / \sum_m |\alpha_m|^2$. For a complete detector this reduces to the standard Born rule. A Gleason-type argument (unitary invariance + additivity) enforces uniqueness in $\dim \geq 3$.

5 New Predictions

Coupling-Induced Tempo Dilation (CITD). In a network of coherently locked oscillators, the intrinsic tempo of a “crowded” leader red-shifts as $\Delta\omega_0 \propto -\sum_j K_{0j}$, mapping to the engineering form $\Delta f/f \approx -\alpha (NgB)/Q_{\text{eff}}$. One-way taps of equal power constitute a null control (no shift).

Coherence-Resource Trade-off. In three-mode interferometry (or three-qubit settings), visibilities obey $V_{AB}^2 + V_{AC}^2 \leq V_0^2$ at fixed source coherence; increasing coherent coupling to C continuously reduces $A-B$ visibility.

6 Failure Conditions (Falsifiability)

Energy or norm non-conservation occurs only if a non-Hermitian $\hat{\Omega}_g$ or asymmetric W is introduced without explicit reservoirs; or if measurement rules violate unitary invariance/additivity; or if nonrelativistic corners are used outside their domain. Empirically, a null in CITD with tight controls or lack of visibility trade-off under coherent coupling would falsify the framework.

7 Experimental Outlook

We outline two precision tests: (i) optical or microwave clock pair with a bidirectional PLL fanout against a one-way power-tap control; (ii) three-path interferometer with tunable coherent coupling. Standard beat-note/visibility analysis suffices to resolve predicted effects (Hz-level in optical clocks).

8 Discussion and Conclusions

Within a single Hermitian action we unify classical, quantum, and relativistic behaviors as limits of a process-based dynamics. Measurement probabilities arise from a resource extremum, not as an axiom. The framework is conservative, covariant in appropriate corners, and—crucially—falsifiable via metrology. Whether nature uses this description is an empirical question, answerable by the proposed tests.

Acknowledgments. The author thanks the GPT-5 computational language model for assistance in formal derivations, organization, and manuscript preparation.

References